

Liming alone does not guarantee the success of sugar maple regeneration

Jean-David Moore ^{*}, Rock Ouimet

Direction de la recherche forestière, Forêt Québec, ministère des Ressources naturelles et des Forêts, 2700 rue Einstein, Sainte-Foy, Québec G1P 3W8, Canada

ARTICLE INFO

Keywords:

Forest liming
Acer saccharum
Dieback
Regeneration
Nutrition

ABSTRACT

Sugar maple (SM; *Acer saccharum* Marshall) is considered to have a high economical and ecological value in northeastern North America. Over the last few decades, dieback and regeneration failure have posed a threat to its sustainability, particularly in acidic and base-poor soils. We evaluated the status of lower regeneration and dominant regeneration in a base-poor northern hardwood stand in Québec, 20 years after a liming treatment. Liming radically increased the density of lower regeneration and the proportion of SM in this stratum, up to levels more than 7 times those of control plots for the highest doses. Dominant regeneration consisted of only 4 % SM (all in the limed plots), while 85 % was American beech (AB; *Fagus grandifolia* Ehrh.). However, liming had no statistically significant effect on the composition of dominant regeneration, and no effect or a negative effect on the diameter and height of SM and AB regeneration. This can probably be explained by competition for light and nutrients resulting from a greater regeneration density and canopy closure in response to lime application. Results show that 20 years after its application in an acidic and base-poor northern hardwood stand, a single lime addition can contribute to increasing the proportion of SM at the lower regeneration stage. However, silvicultural treatments must be considered to ensure that SM lower regeneration reaches a more advanced stratum in the stand.

1. Introduction

Sugar maple (SM; *Acer saccharum* Marshall) is a highly valued forest tree in northeastern North America. It produces both maple syrup and lumber, and its colorful autumn leaves attract tourists. SM stands also have high ecological value due to their contribution to biodiversity. However, over the last few decades, SM has suffered decline and dieback in some regions of the northeastern United States and eastern Canada, as observed by Watmough (2010), Bal et al. (2015), Long et al. (2015) and Moore et al. (2015). Although factors such as insect defoliation, extreme climatic events and climate change have been mentioned as possible causes of SM dieback (Bal et al., 2015; Oswald et al., 2018), the most common factor cited in scientific studies is linked to a low base cation status in soil due in part to soil acidification from past acid deposition (Moore et al., 2021). This puts at risk the sustainability of the species and affects the sustainability of regional industries (Caputo et al., 2016; Beier et al., 2017).

SM is sensitive to both soil acidity and calcium (Ca) deficiency (Wilmot et al., 1995; Duchesne et al., 2002; Bailey et al., 2005; Long et al., 2009; Moore and Ouimet, 2021). Poor soil conditions can lead to growth reduction and decline of mature SM trees (Huggett et al., 2007;

Watmough, 2010; Long et al., 2015; Moore and Ouimet, 2021), as well as to regeneration failure of SM (Kobe et al., 2002; Juice et al., 2006; Long et al., 2011; Moore et al., 2012; Sullivan et al., 2013). These problems could lead to a shift in forest composition toward more acidic and base-poor tolerant species, such as American beech (AB; *Fagus grandifolia* Ehrh.) (Duchesne et al., 2005; Duchesne and Ouimet, 2008; Park and Yanai, 2009; Moore et al., 2012; Marlow and Peart, 2014; Gauthier et al., 2015; Caputo et al., 2016; Pontius et al., 2016). Several studies on the effects of Ca-amendment on mature SM have shown that this scenario can be reversed in acid and base-poor soils by applying the right product at the right rate (Wilmot et al., 1996; Moore and Ouimet, 2006, 2010, 2021; Juice et al., 2006; Huggett et al., 2007; Ouimet et al., 2008; Long et al., 2011; Moore et al., 2012). However, few studies have investigated the long-term effects of Ca addition on SM regeneration (Long et al., 1999: 12 years; Moore et al., 2008: 10 years; Moore et al., 2012: 15 years).

SM regeneration can be affected by factors other than soil limitations, such as browsing, pests and pathogens (Cleavitt et al., 2011; Bose et al., 2018), life history strategies of SM and AB, etc. For example, selective browsing of SM by ungulates may promote AB regeneration (Bose et al., 2017). Moreover, in stands affected by beech bark disease

^{*} Corresponding author.

E-mail address: jean-david.moore@mrnf.gouv.qc.ca (J.-D. Moore).

<https://doi.org/10.1016/j.foreco.2025.122809>

Received 13 January 2025; Received in revised form 12 May 2025; Accepted 13 May 2025

Available online 21 May 2025

0378-1127/© 2025 Elsevier B.V. All rights reserved, including those for text and data mining, AI training, and similar technologies.

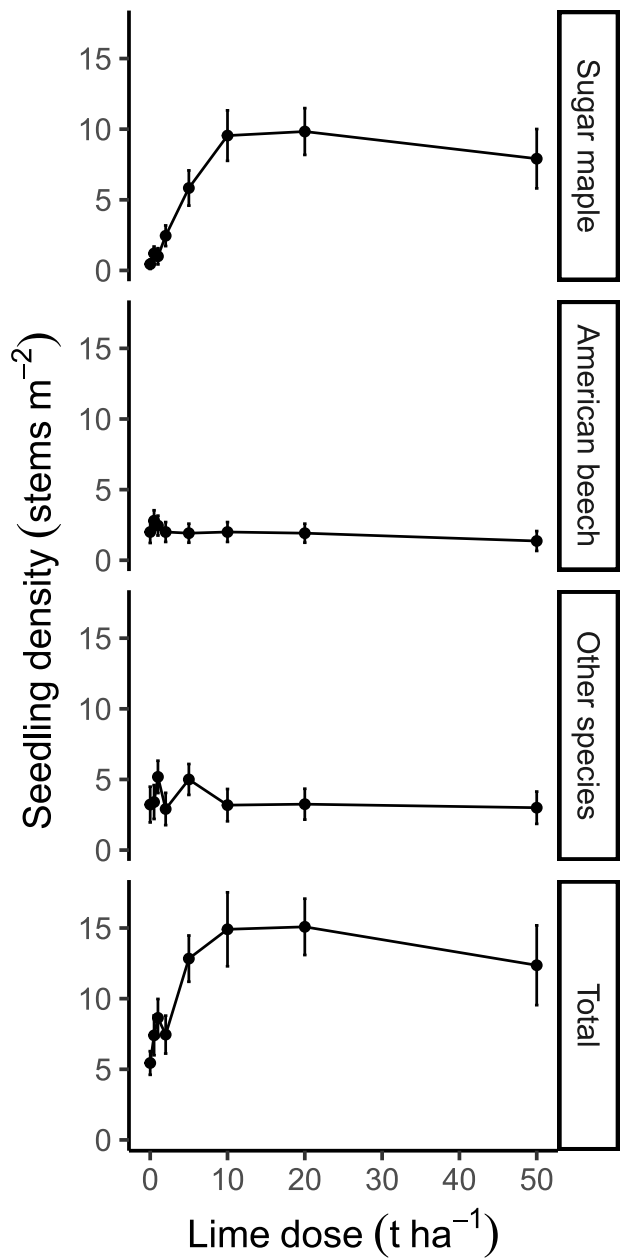


Fig. 1. Adjusted means of the density of lower regeneration as a function of lime rate for sugar maple, American beech, other species, and all species pooled together in the Duchesnay Experimental Forest. Error bars represent $\pm$ SE.

(and/or beech leaf disease), the decline of older beech individuals can initiate a pulse of beech sprouting and cause a huge increase in AB sapling density (Hane, 2003). However, as of today, the main limitation of SM regeneration in the study area relates to the base-poor soil (see below). In this context, the goal of this study was to document the long-term effects of liming (after a 20-year period) on the composition of lower regeneration and advanced regeneration in a base-poor and declining northern hardwood area.

2. Materials and methods

2.1. Site description

The experimental stand is in the vicinity of the Lake Clair watershed at the Duchesnay Experimental Forest, approximately 50 km northwest of Québec City (Québec, Canada; 46° 57' N, 71° 40' W; elevation:

Table 1

Statistical results (ANOVA p -values) 20 years after liming on regeneration density, proportion, diameter, and height of sugar maple (SM), American beech (AB), other species, and for all species pooled together in the Duchesnay Experimental Forest.

Stratum	Variable	Species	Effect of lime treatment		
			Pr>F	Contrasts	
				Linear	Quadratic
Lower regeneration	Density	SM	<0.001	0.004	<0.001
		AB	0.931	0.239	0.864
		Other species	0.716	0.469	0.903
		All species	<0.001	0.117	0.006
	Proportion	SM	<0.001	<0.001	<0.001
		AB	0.091	0.060	0.081
		Other species	0.035	0.101	0.016
	Diameter	SM	0.287	0.300	0.772
		AB	0.330	0.342	0.100
		Other species	0.227	0.563	0.160
		All species	0.032	0.129	0.015
	Height	SM	0.126	0.044	0.336
		AB	0.044	0.209	0.012
		Other species	0.011	0.012	0.008
		All species	0.021	0.013	0.015
Dominant regeneration	Proportion	SM	0.696	0.467	0.373
		AB	0.880	0.552	0.932
		Other species	0.693	0.141	0.375
	Diameter	SM	0.808	0.305	0.670
		AB	0.002	0.061	0.577
		Other species	0.326	0.217	0.271
		All species	0.093	0.063	0.141
	Height	SM	0.734	0.156	0.433
		AB	0.005	<0.001	0.018
		Other species	0.205	0.044	0.785
		All species	0.033	0.024	0.035

270–390 m; average slope: 10 %; mean annual temperature: 3.4 °C; mean annual precipitation: 1300 mm). The vegetation is dominated by SM, accompanied by yellow birch (*Betula alleghaniensis* Britton) and AB. According to the Canadian system of soil classification (Soil Classification Working Group, 1998), the soil is classified as a stony, sandy loam Orthic Ferro-Humic Podzol. The humus is of moder type, and the surface deposit is a very acidic and stony glacial till derived from the granitic gneiss bedrock of the Canadian Shield.

2.2. Stand conditions and disturbance history

Like other watersheds in northeastern North America, Lake Clair was affected by acid deposition until the end of the 1990s. This led to acidic soil and significant net losses of Ca (Watmough et al., 2005; Hazlett et al., 2020).

Before liming in 1994 and until 2014, SM nutritional status revealed severe Ca and Mg deficiencies, particularly for Ca (Moore et al., 2000, 2012; Moore and Ouimet, 2006, 2021), among the worst ever observed for SM (cf. Moore and Ouimet, 2010). The limited availability of Ca can be attributed to the past high levels of acid deposition and the associated Ca leaching, the poor Ca content of the bedrock, and the low rates of mineral weathering in the soil (Houle et al., 1997; Ouimet and Duchesne, 2005).

From 1995–2014, no severe insect defoliation, frost or ice damage

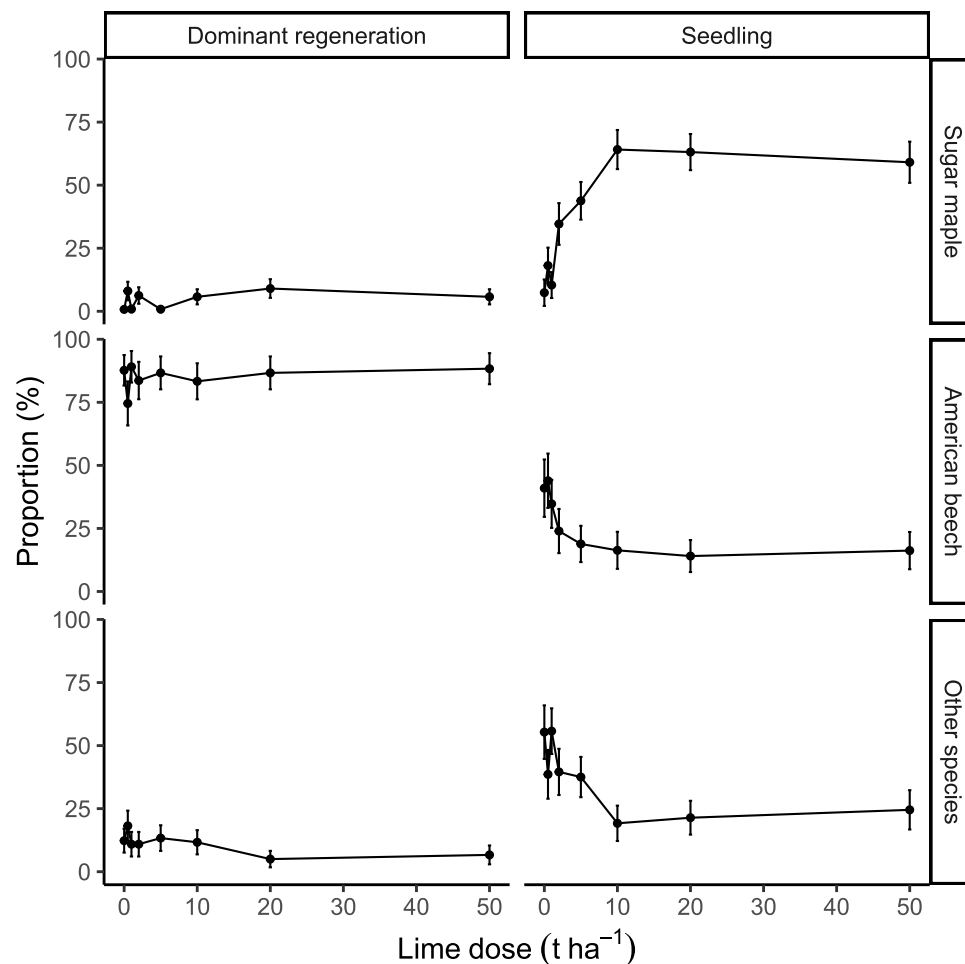


Fig. 2. Adjusted means of the proportion of lower and dominant regeneration represented by sugar maple, American beech, and other species in the Duchesnay Experimental Forest, as a function of lime rate. Error bars represent $\pm$ SE.

was observed in the area. Relatively brief droughts occurred during the summers of 1995, 2002, 2010 and 2012, but they did not result in lasting growth reductions (Moore and Ouimet, 2021). The last forest cutting in the experimental area was a thinning in the 1940s. No sign of beech bark disease or of strong deer browsing has been reported in this area.

2.3. Experimental design

Before treatment, 98 apparently healthy dominant or codominant SM trees spaced at least 15 m apart were selected over a 20 ha area. In 1994, their diameter at breast height was $30.9 \text{ cm} \pm 5.6 \text{ cm}$ (mean $\pm$ SE; range: 20.0–44.0 cm). Before the liming treatment, the entire understory had been cut and removed over a 5 m radius around each selected SM, to reduce the variability of understory abundance and to ensure that lime was spread evenly. There were no other tree stems within this 5 m radius. Trees were numbered and treated randomly in the fall of 1994 (14 replicates for the untreated control, and 12 replicates for each of the 7 liming doses: 0.5, 1, 2, 5, 10, 20 and 50 t ha^{-1}). Dolomitic lime (agricultural grade $\text{CaMg}(\text{CO}_3)_2$, containing 22 % of Ca and 12 % of Mg) was applied manually during the last week of August 1994 within the 5-m cleared radius. More details on the study can be found in Moore et al. (2008). Twenty years later, in the summer of 2014, 2 inventories were conducted to evaluate the response of 2 regeneration stages to liming. First, individuals within the lower regeneration were numbered, and their diameter (15 mm above the forest floor surface) and height were measured in 4 circular subplots of 0.20 m^2 (radius: 25 cm) located at cardinal points 2.5 m from the center of each tree; in total, these 4

subplots cover 0.80 m^2 , or 1 % of the total plot area. In this inventory, species other than SM and AB mainly included (in decreasing order) hobblebush (*Viburnum lantanoides*), striped maple (*Acer pensylvanicum*) and mountain maple (*Acer spicatum*). The second inventory was conducted on the 5 tallest regeneration stems (hereafter called “dominant regeneration”) in each plot (5 m radius), for which species, height and stem diameter 15 cm above the forest floor surface were measured.

2.4. Statistical analyses

We examined the effects of lime treatment as a factor with 8 levels, for 3 groups of species (SM, AB, and others) as well as for all species together. We conducted analyses of variance (ANOVA) on density of lower regeneration, height of both lower and dominant regeneration, and diameter of both lower and dominant regeneration, using the GLIMMIX procedure (SAS Institute, 2002), assuming a normal distribution. A log transformation was used for height and diameter of lower regeneration, and residual variance was estimated by treatment level for SM and total density of lower regeneration. We also analyzed the effects of lime rate on the species composition of lower and dominant regeneration using the LOGISTIC procedure, assuming a binomial distribution with a logit link function, since the data were species proportion (number of stems) of a SM, AB and other species over total number of stems in the lower or in the dominant regeneration. Since all logistic models showed overdispersion (except for the proportion of SM in dominant regeneration), a dispersion parameter was estimated using option scale=williams. In addition, for all analyses, we performed

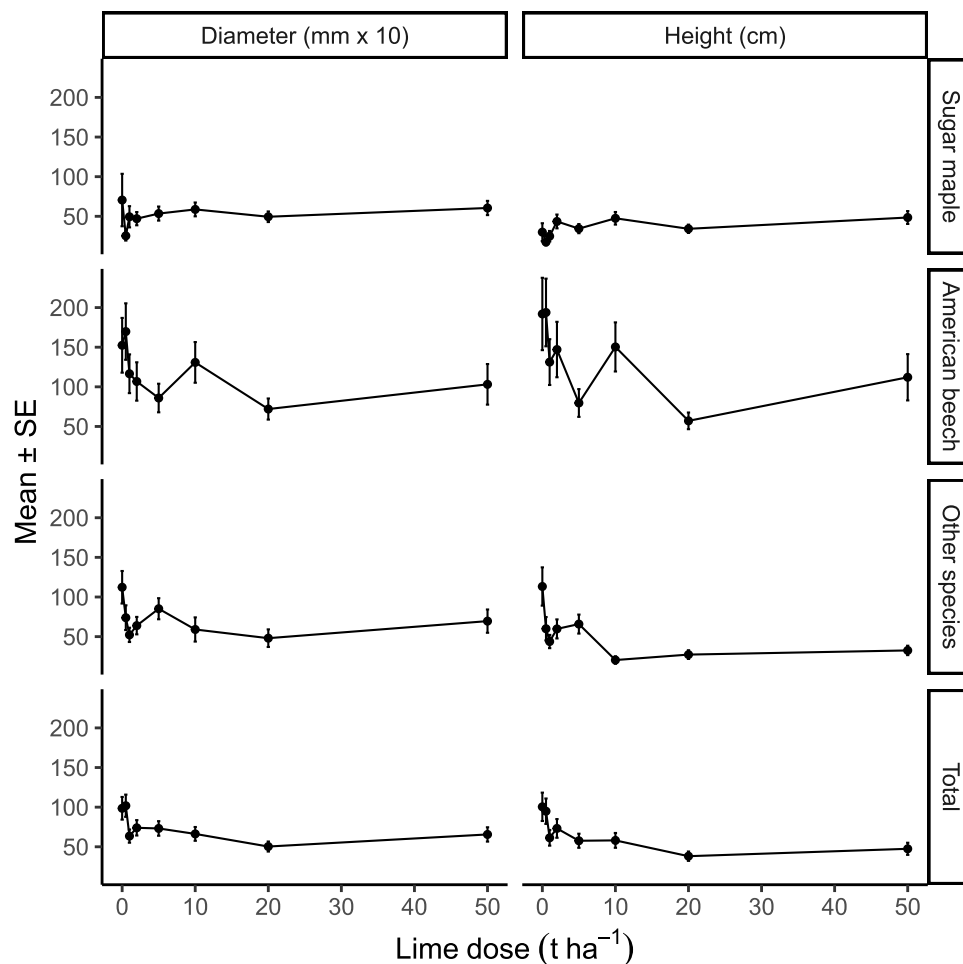


Fig. 3. Adjusted means of the stem diameter and height of lower regeneration represented by sugar maple, American beech, other species, and all species pooled together in the Duchesnay Experimental Forest, as a function of lime rate. Error bars represent $\pm$ SE.

orthogonal polynomial contrasts to separate the linear and quadratic components of the trends across lime treatments.

3. Results

3.1. Regeneration density and proportion

Twenty years after liming, mean density of lower regeneration for SM and AB in the control plots were 0.4 and 2.0 stems·m⁻², respectively, and 5.4 and 2.1 stems·m⁻² in the limed plots, respectively (Fig. 1). Total density of lower regeneration as well as SM density in the lower regeneration both increased with lime rate ($p \leq 0.006$, quadratic effect; Table 1; Fig. 1). No effect was observed for AB and other species ($p \geq 0.239$). SM density increased up to 22 times in the high dose liming treatments (10–50 T·ha⁻¹), and 13 times in the 5 T·ha⁻¹ treatment. The proportion of SM in the lower regeneration cohort was positively influenced by lime rate ($p < 0.001$, quadratic effect; Table 1; Fig. 2), while the proportion of other species decreased ($p = 0.016$, quadratic effect). The proportion of SM in the lower regeneration increased up to 7–8 times for the higher doses, and 5 times for the 5 T·ha⁻¹ treatment. While the proportion of AB decreased by more than 60 % in the higher-dose treatments, no significant effect was observed for this species ($p \geq 0.060$).

Among the dominant regeneration (Fig. 2), 85 % of the stems measured were AB (400 stems), 4 % were SM (18 stems) and 11 % were other species (52 stems). Lime rate had no effect on species composition in the dominant regeneration ($p \geq 0.141$; Table 1; Fig. 2). However, the

measured SM in the dominant regeneration were all located within the limed plots.

3.2. Regeneration height and diameter

Mean diameter of SM and AB lower regeneration in the control plots was 7.1 mm and 15.2 mm, respectively, and 4.9 mm and 11.2 mm in all limed plots, respectively (Fig. 3). Mean height of SM and AB in control plots was 30.0 cm and 191.9 cm, respectively, and 35.8 cm and 124.5 cm in all limed plots, respectively. Lime rate had a negative effect on stem diameter and height of lower regeneration, but only when all species were considered together ($p = 0.015$; quadratic effect; Table 1). No significant effect of lime rate was observed on diameter of lower regeneration for SM, AB, or other species taken individually ($p \geq 0.010$). Height of lower regeneration for AB and other species was negatively affected by lime rate ($p \leq 0.012$, quadratic effect; Table 1). No effect was observed for SM.

The mean diameter of AB dominant regeneration was 36.2 mm in control plots (SM was absent from dominant regeneration in control plots; Fig. 4) and 28.0 mm in all liming treatments (19.4 mm for SM dominant regeneration). The mean height of AB dominant regeneration was 452 cm in control plots and 379 cm in all liming treatments (242 cm for SM dominant regeneration). For dominant regeneration of all species taken together, height decreased after liming ($p = 0.035$, quadratic effect), but not diameter ($p \geq 0.063$; Table 1). For AB and other species, height of dominant regeneration decreased after liming ($p \leq 0.044$, linear effect), but no effect was observed for SM dominant regeneration.

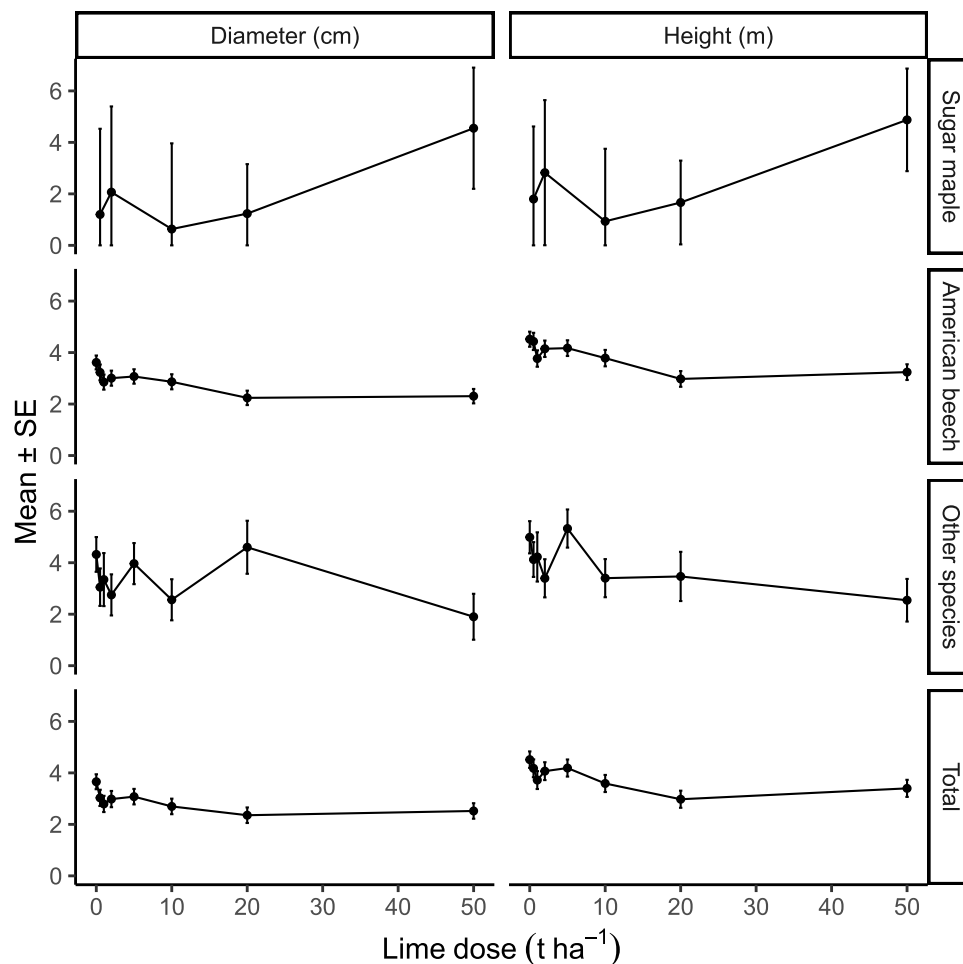


Fig. 4. Adjusted means of the stem diameter and height of dominant regeneration represented by sugar maple, American beech, other species, and all species pooled together in the Duchesnay Experimental Forest, as a function of lime rate. Error bars represent $\pm$ SE.

4. Discussion

After 20 years, liming still had a beneficial effect on the density and proportion of SM lower regeneration in this ecosystem with acid and base-poor soil (Table 1; Figs. 1, 2). This is in line with results from previous measurements in this experiment (10 years after liming: Moore et al., 2008; 15 years after liming: Moore et al., 2012), as well as those of other SM regeneration studies elsewhere in northeastern North America (Long et al., 1999; Juice et al., 2006; Zaccherio and Finzi, 2007). These observations add to the growing body of evidence that Ca may limit the survival of SM regeneration in forests with low soil Ca (Cleavitt et al., 2011; Gauthier et al., 2015). Additionally, it has been demonstrated that adding Ca can help to establish a larger SM seed bank (Cleavitt and Fahey, 2017).

Interestingly, after 20 years, the favorable effect of lime application on SM dominant regeneration disappeared in our experiment (Fig. 2). This contrasts with 15-year results (Moore et al., 2012), which showed a positive effect of liming on the proportion of SM dominant regeneration. This discrepancy could be due to the variation in sample size between the measurements taken in 2009 (10 stems per plot; Moore et al., 2012) and in 2014 (5 stems per plot; this study). This change in sample size reveals that most of the tallest (most advanced) stems in this experiment were AB, which represented 75 %–89 % of the dominant regeneration in both control and limed plots (85 % overall), while SM only represented 0 %–9 % of the dominant regeneration (4 % overall; Fig. 2). Moreover, the diameter and height of AB dominant regeneration were 50 % greater than those of SM, both in control and limed treatments. For lower

regeneration, the difference was even more striking, with AB having a whopping twice the diameter and 3 times the height of SM.

The life history strategies of AB vs. SM could explain the change in AB dominance. Although it is well known that both species are shade-tolerant, AB tends to be more opportunistic than SM under low light conditions, due to its ability to expand vertically and horizontally (Poulson and Platt, 1996). Collin et al. (2017) showed that low light availability associated with the presence of AB was the primary factor leading to SM regeneration failure in the lower Laurentians of Québec, and that allelopathy under AB could also be involved.

In summary, after 20 years, liming had no effect or a negative effect on the height and diameter of lower regeneration and dominant regeneration (Figs. 3, 4; Table 1). This trend had also been observed 15 years after liming (Moore et al., 2012). These findings can likely be attributed to the interaction of two primary factors: first, the intensified competition for light and soil nutrients due to the increased density of regenerating stems, which is a consequence of higher lime application rates; second, the decreased light availability as crown dieback lessens and canopy closure improves after liming (Moore and Ouimet, 2021).

Our findings also highlight the remarkable superiority of AB over SM in this ecosystem. Liming, while enhancing soil fertility and canopy closure after dieback reduction (Moore and Ouimet, 2021), does not affect the dominance of AB. The increased abundance of pole-sized AB has already been documented in this ecosystem (Duchesne et al., 2005). Other studies in SM stands elsewhere in northeastern North America have also reported a shift in forest dynamics in favor of AB (Bose et al., 2017), which tolerates lower levels of Ca and higher soil acidity

(Duchesne and Ouimet, 2009).

In our study, other factors such as browsing preference and beech bark disease do not seem to have influenced this shift in forest dynamics. However, soil acidification and Ca depletion have been identified as important drivers of the composition shift from SM to AB in the Duchesnay Forest and in other SM stands (Jenkins, 1997; Kobe et al., 2002; Duchesne et al., 2005; Bigelow and Canham, 2007; Duchesne and Ouimet, 2009). This shift in composition poses a serious threat to SM sustainability in these ecosystems. In some regions, it could even endanger economic activities linked to SM.

Our results also indicate that liming alone is not enough to promote SM lower regeneration and to enable it to reach the dominant regeneration and, ultimately, the mature tree stages. At a nearby site, Duchesne et al. (2013) showed that the combination of liming and cleaning around SM saplings was the only treatment that provided a competitive advantage to SM over uncleared AB saplings. Moreover, recent studies advocate for changes in management strategies to further promote SM regeneration, where AB is a strong competitor (Henry et al., 2021; St-Jean et al., 2021; Henry and Walters, 2023). For example, St-Jean et al. (2021) found that removing a greater proportion of basal area could favor the abundance and growth of SM. In line with this, Henry and Walters (2023) suggested that single-tree selection management for SM may not be sustainable over large portions of the Great Lakes region.

5. Conclusion

This study demonstrates that a single lime application can significantly boost the proportion of SM in the lower regeneration bank of a base-poor northern hardwood stand, 20 years after treatment. However, silvicultural treatments must be considered to control AB regeneration and ensure that SM lower regeneration reaches a more advanced stratum in the stand.

CRedit authorship contribution statement

Moore Jean-David: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ouimet Rock:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was carried out as part of internal project No. 112310063 of the Direction de la recherche forestière (Forest Research Branch) of the ministère des Ressources naturelles et des Forêts du Québec (Canada), headed by Jean-David Moore. We wish to thank Simon Marcouiller and Benoît Toussaint for field assistance, the personnel of the Laboratoire de chimie organique et inorganique for performing the numerous analyses, Isabelle Auger for help with statistical analysis, and Denise Tousignant for English editing.

Data availability

The authors do not have permission to share data.

References

Bailey, S.W., Horsley, S.B., Long, R.P., 2005. Thirty years of change in forest soils of the Allegheny Plateau, Pennsylvania. *Soil Sci. Soc. Am. J.* 69 (3), 681–690. <https://doi.org/10.2136/sssaj2004.0057>.

- Bal, T.L., Storer, A.J., Jurgensen, M.F., Doskey, P.V., Amacher, M.C., 2015. Nutrient stress predisposes and contributes to sugar maple dieback across its northern range: a review. *Forestry* 88 (1), 64–83. <https://doi.org/10.1093/forestry/cpu051>.
- Beier, C.M., Caputo, J., Lawrence, G.B., Sullivan, T.J., 2017. Loss of ecosystem services due to chronic pollution of forests and surface waters in the Adirondack Region (USA). *J. Environ. Manag.* 191, 19–27. <https://doi.org/10.1016/j.jenvman.2016.12.069>.
- Bigelow, S.W., Canham, C.D., 2007. Nutrient limitation of juvenile trees in a northern hardwood forest: calcium and nitrate are preeminent. *For. Ecol. Manag.* 243 (2–3), 310–319. <https://doi.org/10.1016/j.foreco.2007.03.027>.
- Bose, A.K., Weiskittel, A., Wagner, R.G., 2017. Occurrence, pattern of change, and factors associated with American beech-dominance in stands of the northeastern USA forest. *For. Ecol. Manag.* 392, 202–212. <https://doi.org/10.1016/j.foreco.2017.03.003>.
- Bose, A.K., Wagner, R.G., Roth, B.E., Weiskittel, A.R., 2018. Influence of browsing damage and overstory cover on regeneration of American beech and sugar maple nine years following understory herbicide release in central Maine. *New For.* 49, 67–85. <https://doi.org/10.1007/s11056-017-9605-5>.
- Caputo, J., Beier, C.M., Sullivan, T.J., Lawrence, G.B., 2016. Modeled effects of soil acidification on long-term ecological and economic outcomes for managed forests in the Adirondack region (USA). *Sci. Total Environ.* 565, 401–411. <https://doi.org/10.1016/j.scitotenv.2016.04.008>.
- Cleavitt, N.L., Fahey, T.J., 2017. Seed production of sugar maple and American beech in northern hardwood forests, New Hampshire, USA. *Can. J. For. Res.* 47 (7), 985–990. <https://doi.org/10.1139/cjfr-2017-0096>.
- Cleavitt, N.L., Fahey, T.J., Battles, J.J., 2011. Regeneration ecology of sugar maple (*Acer saccharum*): seedling survival in relation to nutrition, site factors, and damage by insects and pathogens. *Can. J. For. Res.* 41 (2), 235–244. <https://doi.org/10.1139/X10-210>.
- Collin, A., Messier, C., Kembel, S.W., Bélanger, N., 2017. Low light availability associated with American beech is the main factor for reduced sugar maple seedling survival and growth rates in a hardwood forest of Southern Quebec. *Forests* 8 (11), 413. <https://doi.org/10.3390/f8110413>.
- Duchesne, L., Ouimet, R., 2008. Population dynamics of tree species in southern Quebec, Canada: 1970–2005. *For. Ecol. Manag.* 255 (7), 3001–3012. <https://doi.org/10.1016/j.foreco.2008.02.008>.
- Duchesne, L., Ouimet, R., 2009. Present-day expansion of American beech in northeastern hardwood forests: does soil base status matter? *Can. J. For. Res.* 39 (12), 2273–2282. <https://doi.org/10.1139/X09-172>.
- Duchesne, L., Ouimet, R., Houle, D., 2002. Basal area growth of sugar maple in relation to acid deposition, stand health, and soil nutrients. *J. Environ. Qual.* 31 (5), 1676–1683. <https://doi.org/10.2134/jeq2002.1676>.
- Duchesne, L., Ouimet, R., Moore, J.-D., Paquin, R., 2005. Changes in structure and composition of maple-beech stands following sugar maple decline in Québec, Canada. *For. Ecol. Manag.* 208 (1–3), 223–236. <https://doi.org/10.1016/j.foreco.2004.12.003>.
- Duchesne, L., Moore, J.-D., Ouimet, R., 2013. Partitioning the effect of release and liming on growth of sugar maple and American beech saplings. *North. J. Appl. For.* 30 (1), 28–36. <https://doi.org/10.5849/njaf.12-012>.
- Gauthier, M.-M., Guillemette, F., Bédard, S., 2015. On the relationship between saplings and ingrowth in northern hardwood stands. *For. Ecol. Manag.* 358, 261–271. <https://doi.org/10.1016/j.foreco.2015.09.020>.
- Hane, E.N., 2003. Indirect effects of beech bark disease on sugar maple seedling survival. *Can. J. For. Res.* 33 (5), 807–813. <https://doi.org/10.1139/x03-008>.
- Hazlett, P.W., Emilson, C.E., Lawrence, G.B., Fernandez, I.J., Ouimet, R., Bailey, S.W., 2020. Reversal of forest soil acidification in the northeastern United States and eastern Canada: site and soil factors contributing to recovery. *Soil Syst.* 4 (3), 54. <https://doi.org/10.3390/soilsystems4030054>.
- Henry, C.R., Walters, M.B., 2023. Sugar maple (*Acer saccharum*) age structure reveals limited establishment and development of age cohorts in response to selection management in northern hardwood forests. *For. Ecol. Manag.* 546, 121356. <https://doi.org/10.1016/j.foreco.2023.121356>.
- Henry, C.R., Walters, M.B., Finley, A.O., Roloff, G.J., Farinosi, E.J., 2021. Complex drivers of sugar maple (*Acer saccharum*) regeneration reveal challenges to long-term sustainability of managed northern hardwood forests. *For. Ecol. Manag.* 479, 118541. <https://doi.org/10.1016/j.foreco.2020.118541>.
- Houle, D., Paquin, R., Camiré, C., Ouimet, R., Duchesne, L., 1997. Response of the Lake Clair Watershed (Duchesnay, Quebec) to change in precipitation chemistry (1988–1994). *Can. J. For. Res.* 27 (11), 1813–1821. <https://doi.org/10.1139/x97-143>.
- Huggett, B.A., Schaberg, P.G., Hawley, G.J., Eagar, C., 2007. Long-term calcium addition increases growth release, wound closure, and health of sugar maple (*Acer saccharum*) trees at the Hubbard Brook Experimental Forest. *Can. J. For. Res.* 37 (9), 1692–1700. <https://doi.org/10.1139/X07-042>.
- Jenkins, J., 1997. Hardwood regeneration failure in the Adirondacks: Preliminary studies of incidence and severity. Working Paper No. 9. Wildlife Conservation Society, Bronx, New York, p. 67. (<https://library.wcs.org/Scientific-Research/Publication-s-Bibliographies/Working-Papers.aspx>).
- Juice, S.M., Fahey, T.J., Siccama, T.G., Driscoll, C.T., Denny, E.G., Eagar, C., Cleavitt, N.L., Minocha, R., Richardson, A.D., 2006. Response of sugar maple to calcium addition to northern hardwood forest. *Ecology* 87 (5), 1267–1280. [https://doi.org/10.1890/0012-9658\(2006\)87\[1267:ROSMTC\]2.0.CO;2](https://doi.org/10.1890/0012-9658(2006)87[1267:ROSMTC]2.0.CO;2).
- Kobe, R.K., Likens, G.E., Eagar, C., 2002. Tree seedling growth and mortality responses to manipulations of calcium and aluminum in a northern hardwood forest. *Can. J. For. Res.* 32 (6), 954–966. <https://doi.org/10.1139/x02-018>.
- Long, R.P., Horsley, S.B., Lilja, P.R., 1999. Impact of forest liming on growth and crown vigor of sugar maple and associated hardwoods. In: Horsley, S.B., Long, R.P. (Eds.),

- Sugar Maple Ecology and Health. Proceedings of an International Symposium. 2–4 June 1998. USDA For. Serv., Northeastern Research Station, Gen. Tech. Rep. NE-261, Warren, PA, pp. 55–58. <https://research.fs.usda.gov/treesearch/13157>.
- Long, R.P., Horsley, S.B., Hallett, R.A., Bailey, S.W., 2009. Sugar maple growth in relation to nutrition and stress in the northeastern United States. *Ecol. Appl.* 19 (6), 1454–1466. <https://www.jstor.org/stable/40346260>.
- Long, R.P., Horsley, S.B., Hall, T.J., 2011. Long-term impact of liming on growth and vigor of northern hardwoods. *Can. J. For. Res.* 41 (6), 1295–1307. <https://doi.org/10.1139/x11-049>.
- Long, R.P., Bailey, S.W., Horsley, S.B., Hall, T.J., Swistock, B.R., DeWalle, D.R., 2015. Long-term effects of forest liming on soil, soil leachate, and foliage chemistry in northern Pennsylvania. *Soil Sci. Soc. Am. J.* 79 (4), 1223–1236. <https://doi.org/10.2136/sssaj2014.11.0465>.
- Marlow, J., Peart, D.R., 2014. Experimental reversal of soil acidification in a deciduous forest: implications for seedling performance and changes in dominance of shade-tolerant species. *For. Ecol. Manag.* 313, 63–68. <https://doi.org/10.1016/j.foreco.2013.10.036>.
- Moore, J.-D., Ouimet, R., 2006. Ten-year effect of dolomitic lime on the nutrition, crown vigor and growth of sugar maple. *Can. J. For. Res.* 36, 1834–1841. <https://doi.org/10.1139/x06-081>.
- Moore, J.-D., Ouimet, R., 2010. Effects of two Ca fertilizer types on sugar maple vitality. *Can. J. For. Res.* 40 (10), 1985–1992. <https://doi.org/10.1139/X10-136>.
- Moore, J.-D., Ouimet, R., 2021. Liming still positively influences sugar maple nutrition, vigor and growth, 20 years after a single application. *For. Ecol. Manag.* 490, 119103.
- Moore, J.-D., Camiré, C., Ouimet, R., 2000. Effects of liming on the nutrition, vigor and growth of sugar maple at the Lake Clair watershed, Quebec, Canada. *Can. J. For. Res.* 30 (5), 725–732. <https://doi.org/10.1139/x00-009>.
- Moore, J.-D., Duchesne, L., Ouimet, R., 2008. Soil properties and maple-beech regeneration a decade after liming in a northern hardwood stand. *For. Ecol. Manag.* 255 (8–9), 3460–3468. <https://doi.org/10.1016/j.foreco.2008.02.026>.
- Moore, J.-D., Ouimet, R., Duchesne, L., 2012. Soil and sugar maple response 15 years after dolomitic lime application. *For. Ecol. Manag.* 281, 130–139. <https://doi.org/10.1016/j.foreco.2012.06.026>.
- Moore, J.-D., Ouimet, R., Long, R.P., Bukaveckas, P.A., 2015. Ecological benefits and risks arising from liming sugar maple dominated forests in northeastern North America. *Environ. Rev.* 23, 66–77.
- Oswald, E.M., Pontius, J., Rayback, S.A., Schaberg, P.G., Wilmot, S.H., Dupigny-Giroux, L.A., 2018. The complex relationship between climate and sugar maple health: climate change implications in Vermont for a key northern hardwood species. *For. Ecol. Manag.* 422, 303–312. <https://doi.org/10.1016/j.foreco.2018.04.014>.
- Ouimet, R., Duchesne, L., 2005. Base cation mineral weathering and total release rates from soils in three calibrated forest watersheds on the Canadian Boreal Shield. *Can. J. Soil Sci.* 85 (2), 245–260. <https://doi.org/10.4141/S04-061>.
- Ouimet, R., Moore, J.-D., Duchesne, L., 2008. Effects of experimental acidification and alkalization on soil and growth and health of *Acer saccharum* Marsh. *J. Plant Nutr. Soil Sci.* 171 (6), 858–871. <https://doi.org/10.1002/jpln.200700197>.
- Park, B.B., Yanai, R.D., 2009. Nutrient concentrations in roots, leaves and wood of seedling and mature sugar maple and American beech at two contrasting sites. *For. Ecol. Manag.* 258 (7), 1153–1160. <https://doi.org/10.1016/j.foreco.2009.06.003>.
- Pontius, J., Halman, J.M., Schaberg, P.G., 2016. Seventy years of forest growth and community dynamics in an undisturbed northern hardwood forest. *Can. J. For. Res.* 46 (70), 959–967. <https://doi.org/10.1139/cjfr-2015-0304>.
- Poulson, T.L., Platt, W.J., 1996. Replacement patterns of beech and sugar maple in Warren Woods, Michigan. *Ecology* 77 (4), 1234–1253.
- SAS Institute, 2002. SAS version 9. SAS Institute, Cary, North Carolina, USA.
- Soil Classification Working Group, 1998. The Canadian system of soil classification, 3rd edition. Agriculture and Agri-Food Canada Publication No. 1646, Ottawa, Ontario, p. 187. <https://sis.agr.gc.ca/cansis/publications/manuals/1998-cssc-ed3/index.html>.
- St-Jean, É., Meunier, S., Nolet, P., Messier, C., Achim, A., 2021. Increased levels of harvest may favour sugar maple regeneration over American beech in northern hardwoods. *For. Ecol. Manag.* 499, 119607. <https://doi.org/10.1016/j.foreco.2021.119607>.
- Sullivan, T.J., Lawrence, G.B., Bailey, S.W., McDonnell, T.C., Beier, C.M., Weathers, K.C., McPherson, G.T., Bishop, D.A., 2013. Effects of acidic deposition and soil acidification on sugar maple trees in the Adirondack Mountains, New York. *Environ. Sci. Technol.* 47 (22), 12687–12694. <https://doi.org/10.1021/es401864w>.
- Watmough, S.A., 2010. Assessment of the potential role of metals in sugar maple (*Acer saccharum* Marsh) decline in Ontario, Canada. *Plant Soil* 332, 463–474. <https://doi.org/10.1007/s11104-010-0313-6>.
- Watmough, S.A., Aherne, J., Alewell, C., Arp, P.A., Bailey, S.W., Clair, T.A., Dillon, P., Duchesne, L., Eimers, C., Fernandez, I., Foster, N., Larssen, T., Miller, E.K., Mitchell, M., Page, S., 2005. Sulphate, nitrogen and base cation budgets at 21 forested catchments in Canada, the United States and Europe. *Environ. Sci. Technol.* 109, 1–36. <https://doi.org/10.1007/s10661-005-4336-z>.
- Wilmot, T.R., Ellsworth, D.S., Tyree, M.T., 1995. Relationships among crown condition, growth, and stand nutrition in seven northern Vermont sugarbushes. *Can. J. For. Res.* 25 (3), 386–397. <https://doi.org/10.1139/x95-043>.
- Wilmot, T.R., Ellsworth, D.S., Tyree, M.T., 1996. Base cation fertilization and liming effects on nutrition and growth of Vermont sugar maple stands. *For. Ecol. Manag.* 84 (1–3), 123–134. [https://doi.org/10.1016/0378-1127\(96\)03743-7](https://doi.org/10.1016/0378-1127(96)03743-7).
- Zaccherio, M.T., Finzi, A.C., 2007. Atmospheric deposition may affect northern hardwood forest composition by altering soil nutrient supply. *Ecol. Appl.* 17 (7), 1929–1941. <https://doi.org/10.1890/06-2067.1>.